SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 1st November 2026

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|---|
| Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility. |

## 1. Team & Project Overview

| | |
|---|---|
| Team Name | Chill Chill |
| Project / Solution Name | KerjayaKu - AI Career Exploration and Learning Companion |
| Name (Leader) | Edixon Teo Zan Wei |
| Email & Phone (Leader) | edixontzw@gmail.com / 010 880 9196 |
| University Affiliation(s) | Taylor's University |
| Live Prototype URL | https://kerjayaku-demo.onrender.com/ |
| Code Repository URL (public / access granted to judges) | https://github.com/Aloysius8895/xmumHack |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

| |
|---|
| KerjayaKu is a browser-based 3D career and learning prototype. The Three.js/Vite interface communicates with a Node.js REST API for profiles, careers, learning roadmaps, SDGs, pets, reports and rankings. DeepSeek inference extracts skills and produces explanations and learning content. Validated identifiers and deterministic rules control scores, task progress and points; no model is trained or fine-tuned by the team.<br>Market collectors obtain job-ad samples through JSearch or supported public-page sources. They normalise, deduplicate and store observations, then publish dated market snapshots. Public economic and workforce datasets provide additional context and AI-adoption scenario evidence.<br>The game supports file-backed storage, PostgreSQL and an optional AWS persistence adapter. Python FastAPI/SQLite and Telegram integrations support companion conversations and reports; n8n workflow definitions support scheduling and automation. Their deployment depends on configuration. |

Replit configuration builds the frontend and serves it with the Node API and collector. The locally verified AI configuration uses deepseek-flash. RunPod and other AI adapters exist, but code presence does not establish an active deployment. The prototype needs browser GPU support and is not a production multi-tenant service.

## 2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

| Name & Version | Purpose in Project | License |
|---|---|---|
| JavaScript / Node.js 24+; HTML; CSS | Game client, native HTTP API, collector and browser UI | Node.js MIT and bundled notices; web standards |
| Python 3.11 deployment configuration; SQLite | Companion service, data and asset scripts; local database | PSF; SQLite public domain |
| Three.js 0.183.2; Rapier 0.17.3 | 3D rendering with WebGPU/WebGL and physics | MIT; Apache-2.0 |
| Vite 7.2.4; Stylus 0.64.0 | Frontend build and styling | MIT |
| GSAP 3.12.5; Howler 2.2.4 | Animation and audio | GSAP package standard license; Howler MIT |
| @anthropic-ai/sdk 0.135.0; pdf-parse 2.4.5 | Server-side model protocol adapter; CV PDF text extraction | MIT; Apache-2.0 |
| Name & Version | Purpose in Project | License |
| pg 8.23.1; dotenv 17.2.3 | Optional PostgreSQL persistence; server environment variables | MIT; BSD-2-Clause |
| FastAPI >=0.110; Uvicorn >=0.29; HTTPX >=0.27 | Python webhook/API service and HTTP client; requirement ranges | MIT; BSD-3-Clause; BSD-3-Clause |
| Python anthropic >=0.40; openai >=1.50; python-dotenv >=1,<2 | Companion provider adapters and environment handling | MIT; Apache-2.0; BSD-3-Clause |

| | | |
|---|---|---|
| Playwright 1.62.1 Node; Python >=1.40; pytest >=8 | Browser collector/automation and automated checks | Apache-2.0; pytest MIT |
| Beautiful Soup >=4.12; pandas >=2.2; PyYAML >=6 | Public-page parsing, data processing and configuration | MIT; BSD-3-Clause; MIT |
| Blender 5.2.2 LTS documented; glTF Transform CLI 4.1.4 / modules 4.2.1 | 3D asset generation, animation and optimisation | Blender GPL; glTF Transform MIT; asset rights separate |
| n8n workflow definitions; instance version not recorded | Market and messaging workflow automation; activation not verified | n8n Sustainable Use / applicable enterprise terms |
| Bruno Simon folio-2025 foundation | Adapted 3D game foundation; upstream notice retained | MIT; other assets retain their own terms |
| Google Fonts; locally bundled fonts, audio and 3D assets | Nunito, Amatic SC, Fredoka, Noto Sans SC, ZCOOL KuaiLe; game assets | Individual font and asset notices apply; final asset rights review required |

## 3. APIs & Cloud Services

### 3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| DeepSeek API | Locally verified deepseek-flash inference; profile, mentor and learning assistance | Team account; metered API; no official sponsor resource |
| Telegram Bot API | Companion messages, callbacks and learning reports; live availability not verified | Bot API; no sponsor resource |
| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |

| | | |
|---|---|---|
| JSearch via RapidAPI / OpenWeb Ninja | Job-ad collection adapter and refresh pipeline; actual quota depends on credentials | Account subscription/free quota not recorded; no sponsor resource |
| Google Analytics / Google tag | Tag G-JMSN30BQ5J exists in frontend; telemetry execution and account ownership require checking | Standard web tag; plan not recorded |
| Google Fonts CSS/font endpoints | Browser font delivery through fonts.googleapis.com and fonts.gstatic.com | Public font delivery; individual font licenses |
| RunPod vLLM; Anthropic; Replit AI proxy; AWS Bedrock / DeepSeek worker | Alternative AI adapters/earlier configurations; not the locally verified active provider | Provider/account terms; active plans and usage not established |
| Meta WhatsApp Cloud API | Implemented optional companion channel; approval/activation not established | Not claimed active; Meta messaging/account terms |
| Public data and job-source HTTP endpoints | Hugging Face, official statistics, LinkedIn, Indeed and Ricebowl; see Section 4.2 | Public access or configured API; source terms apply |

## 3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit deployment | Game frontend, Node API and collector deployment configuration; final public URL to be supplied | Cloud Run target in .replit; account plan/region not recorded; no sponsor credits used |
| RunPod | Optional vLLM inference path and retained historical proof document; current endpoint not verified | No official SDC provision; exact model, plan and region not recorded |
| GitHub | Source control and repository distribution | Team repository; judge access to be checked |
| PostgreSQL or local JSON / SQLite | Configurable game persistence and companion/collector storage | Provider selected by configuration; local file mode available; managed host not established |

| | | |
|---|---|---|
| n8n | Workflow definitions for market collection and messaging | Self-hosted/cloud instance and region not established |
| Service & Provider | Purpose in Project | Plan / Region |
| AWS API Gateway, Lambda, Aurora PostgreSQL Data API | Optional persistence/compute architecture in repository; not verified deployed | No live plan/region claimed |
| AWS Secrets Manager, S3, CloudWatch, X-Ray | Supporting optional infrastructure: secrets, migration objects and observability | No live plan/region claimed |

# 4. AI Tools, Models & Datasets

## 4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| deepseek-flash (configured API identifier) | DeepSeek official API | Pre-trained inference; local call verified; no team training or fine-tuning | DeepSeek API terms; weights not redistributed |
| Alternative RunPod-hosted model; identifier not recorded | RunPod / vLLM adapter | Optional/previous configuration; current model not verified | Host and underlying model terms must both apply |
| Claude / other configured provider alternatives | Anthropic, Replit proxy, AWS Bedrock adapters | Supported inference paths; not claimed active solely because adapters exist | Respective API terms; exact runtime model not established |
| Karpathy / Rosling perspective cards | Local source-based prompt material | Context for generated responses; not separate models or endorsements | Underlying source rights retained; Nuwa method MIT |

## 4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Career, skills, course and prerequisite catalogues | Repository src/data/config and server reference files | Team-curated runtime reference | Repository terms plus original course/source rights | N for catalogues |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Anthropic Economic Index and O*NET task mappings | Anthropic public releases; d ata/ai-exposure/source-peri ods.json retains exact URLs and periods | Public aggregate/re ference; derived task mappings | Publisher dataset terms; O*NET attribution; verify release-specific notices | N for published aggregates |
| WEF Future of Jobs 2025 | https://www3.weforum.org/ docs/WEF_Future_of_Jobs _Report_2025.pdf | Public report; directional skill signals | WEF publication terms; no open-license assumption | N for aggregate inputs |
| TalentCorp MyCOL and AI Digital Green Economy Impact Study | https://www.talentcorp.com .my/ ; exact publication URLs in data/official-signal s/signals.json | Public reports; occupational signals | Publisher terms and attribution | N for aggregate inputs |
| DOSM and Techmap Malaysia job statistics | https://www.dosm.gov.my/ ; https://jobdatafeeds.com/d ata/countries/my | Public historical ag gregate/reference | Original publisher terms; coverage differs by source | N for aggregate inputs |
| JobStreet 2024 collection azrai99/ job-dataset | https://huggingface.co/datas ets/azrai99/job-dataset | Public third-party job listings; historical aggregates | Current dataset card: Apache-2.0; original listing rights remain relevant | Y possible in source d escriptions |
| JobStreet 2025 collection seemarn/jobstreet | https://huggingface.co/datas ets/seemarn/jobstreet | Public third-party source; derived historical aggregates | License not established; rights confirmation required before redistribution | Y possible in raw source |
| FDE / OpenJobData Jobs-Dataset-v2 | https://huggingface.co/datas ets/scholarweave/Jobs-Data set-v2 | Public job listings; sampled/reference inputs | Dataset card: Apache-2.0; source rights apply | Y possible in raw source |

| | | | | |
|---|---|---|---|---|
| JSearch, LinkedIn, Indeed and Ricebowl observations | https://www.linkedin.com/j obs/ ; https://malaysia.indee d.com/ ; https://www.ricebowl.my/ ; JSearch | Self-collected third-party samples; deduplicated snapshots | Platform/API terms; no blanket redistribution permission claimed | Y possible in source contacts |
| UN SDG goals and targets | https://sdgs.un.org/goals ; repository goal/target mappings | Public taxonomy; team-derived career links | UN content/logo rules; no UN endorsement | N |
| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
| Synthetic profiles, job descriptions, reports and mock market/exposure data | Repository tests and data/mock | Synthetic testing/demo; not measured outcomes | Team fixtures; exclude synthetic statistics from impact claims | N if synthetic |
| User CVs, skill profiles, chats, learning reports and identifiers | Voluntary in-app uploads/input; Telegram messages where enabled | Private user data; runtime inference and progress | User permission and provider terms; no model-training rights assumed | Y; CVs may be sensitive |
| Public expert research and perspective summaries | src/expert source/reference folders | Public-source prompt context; no training | Original source rights; Nuwa method MIT | Y: public n ames/biogr aphy |

## 4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

[X] Yes [ ] No

If yes, list them below:

| Tool & Version | Used For | Extent of Use (minor / moderate / major) |
|---|---|---|
| Claude; exact model/session versions not recorded | Team-confirmed AI assistance during development; review of code/content | Major (development assistance) |

| | | |
|---|---|---|
| Kilo Code; extension/model versions not recorded | Team-confirmed coding assistant; underlying model selection not established | Moderate (coding assistance) |
| OpenAI Codex / ChatGPT coding environment; model versions not recorded | Repository analysis, code changes/debugging, testing assistance and documentation | Major (coding and documentation) |
| Built-in OpenAI imagegen tool; exact model not recorded | Fantasy interface frame generation/edits documented in static/interface/README.md; retained asset, not current panel renderer | Moderate (asset-specific) |
| Tool & Version | Used For | Extent of Use (minor / moderate / major) |
| Nuwa distillation method; repository skill/reference material | Public-source expert perspective research and prompt-card preparation | Minor (prompt/reference method; not a runtime model) |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted output is reviewed against source code, catalogue identifiers and referenced datasets. The repository includes unit, integration, browser/workflow checks and production-build commands. Model output is validated before applying game rules; users can edit extracted skills. We distinguish public reference data, synthetic fixtures and collected market observations. The final deployment, third-party asset permissions and all members' tool histories still require the leader's review before signing. The team remains responsible for submitted code, content and declarations; generated text is not treated as independently verified evidence.

## 5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

[ ] Yes [X] No

If yes, list them below:

| Sponsor | Resource (API / credits / platform / data) | How It Was Used in the Project |
|---|---|---|
| None | No official SDC sponsor-provided resources used, as confirmed by the team. | Not applicable. Commercial platforms or API accounts are not described as sponsored resources. |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Not applicable. The team confirmed that no official SDC sponsor-provided resource was used. Replit hosting configuration, DeepSeek inference and optional RunPod integration do not by themselves establish sponsorship. The application separates provider adapters, deterministic rules and persistence, allowing a host or model API to be replaced with configuration and integration work. Any replacement would need cost, privacy and behaviour checks.

# 6. Security, Privacy & Responsible Use

## 6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State “No user data is collected” if applicable.

[Maximum 150 words]

The application can collect display names, CV text, skill profiles, target job descriptions, chat messages and learning reports. Relevant content is sent to the configured AI provider; Telegram processes messages and identifiers when its integration is enabled. Google Fonts requests and the embedded Google Analytics tag may also transmit browser/network metadata.
Game state uses JSON or configured PostgreSQL/AWS storage; companion and collector components use SQLite. Credentials are server-side environment variables, excluded from this document. Integration routes include tokens/secrets and duplicate-message checks. Local file mode is not production account isolation. Records persist until reset, deletion or operator cleanup; some histories have count limits, but there is no complete automatic retention schedule. External-provider retention is separate. Demonstrations should use synthetic records. Real CV use requires informed consent, an explicit deletion policy and access-control review.

## 6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

AI may misinterpret skills, generate inaccurate advice or misclassify a report. Identifier checks, constrained catalogues, editable profiles and deterministic progress rules reduce errors. Reported learning does not independently prove competence, employability or real-world SDG impact.
Market samples are incomplete and source-dependent. AEI reflects observed Claude usage, not job-loss probability; mapped tasks and team-set scenario thresholds involve assumptions. Synthetic history must not be presented as measured data. Some dataset and bundled-asset permissions still need verification.
API outages, quotas and stale snapshots affect availability. Optional adapters and workflow definitions do not guarantee live services. The 3D interface depends on GPU/browser capabilities and may perform poorly on weaker devices. Cross-platform operation requires testing on target browsers; no crash-free guarantee is made. Production use requires stronger multi-user identity controls, retention rules and deployment monitoring.

## 7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

| | |
|---|---|
| [ ] | All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.<br>Status: Pending leader review: this document lists confirmed tools and repository-supported alternatives; historical model/account details are not all recorded. |
| [ ] | All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.<br>Status: Pending: verify third-party assets, job-source reuse permissions and missing source licenses before certifying. |
| [ ] | No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.<br>Status: Pending final secret audit: this document contains no credentials; repository history and production bundle are not certified secret-free. |
| [ ] | Any use of generative-AI tools for development has been disclosed and reviewed by the team.<br>Status: Pending leader review: Claude, Kilo Code, Codex and documented image generation are disclosed; final team sign-off remains required. |
| [X] | Any use of official SDC sponsor resources has been declared in Section 5.<br>Status: Confirmed: no official SDC sponsor-provided resources used (Section 5). |
| [ ] | The prototype described here is accessible to judges through the links provided in Section 1.<br>Status: Pending: insert final Demo URL and verify judge access to prototype and repository. |
| [ ] | All fields in this template have been completed within the word limits stated.<br>Status: Pending final submission: narrative word limits met; deployment URL, signing date and leader signature remain to be completed. |

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: ____ Edixon Teo Zan Wei____

Date signed: _____1 November 2026_____

## Appendix A Implementation and source notes

Reviewed revision: f5c1ebf on main, pulled on 1 November 2026. Package versions below are lockfile versions, not a certification of the deployed environment. Python manifests specify ranges. No new performance, deployment or full-suite result is claimed by this declaration.

Main flow: browser UI -> Node.js API -> validated model response + rules -> persisted progress. Market flow: sources -> collect -> normalise/deduplicate -> dated snapshots -> trends/scenario evidence. Companion flow, when enabled: Telegram -> FastAPI/n8n -> game integration API -> stored report and reply.

Runtime evidence: .replit; src/game/server; src/game/package-lock.json; src/game/server/persistence.js; docs/SYSTEM.md; docs/ROADMAP-INPUTS.md. AI-adoption release periods and source URLs: data/ai-exposure/source-periods.json. Official labour signals: data/official-signals. Optional AWS infrastructure is not evidence of deployment.

Upstream foundation: Bruno Simon, https://github.com/brunosimon/folio-2025 . Upstream MIT notice must remain. The root Apache-2.0 license and package metadata do not replace individual third-party licenses. Fonts, audio, textures, logos and imported/generated characters need their own provenance and permission review.

The Google Analytics tag is declared because it remains in the HTML, not because analytics operation has been verified. The team should verify ownership, consent behaviour and necessity before a public release.

Current dataset cards identify azrai99/job-dataset and scholarweave/Jobs-Dataset-v2 as Apache-2.0 (checked 1 November 2026). No open license is assumed for seemarn/jobstreet. Dataset licensing does not automatically settle original job-ad, contact-data or platform rights.

Course links point to external resources rather than transferring course ownership or certifying eligibility. Examples include Select Star SQL, Kaggle Learn, OpenIntro, Python documentation and other catalogue links. Any HRD Corp claim must be checked against the actual course/provider listing.

## Appendix B Direct JavaScript dependencies

| Package | Locked version | License recorded in package/lock |
|---|---|---|
| @anthropic-ai/sdk | 0.135.0 | MIT |
| @dimforge/rapier3d | 0.17.3 | Apache-2.0 |
| @gltf-transform/cli | 4.1.4 | MIT |
| @gltf-transform/core | 4.2.1 | MIT |
| @gltf-transform/extensions | 4.2.1 | MIT |
| @gltf-transform/functions | 4.2.1 | MIT |
| @tweakpane/plugin-camerakit | 0.3.0 | MIT |
| @tweakpane/plugin-essentials | 0.2.1 | MIT |
| @vitejs/plugin-basic-ssl | 2.1.0 | MIT |
| camera-controls | 3.1.2 | MIT |
| dotenv | 17.2.3 | BSD-2-Clause |
| emoji-regex | 10.4.0 | MIT |
| glob | 13.0.0 | BlueOak-1.0.0 |
| gsap | 3.12.5 | GSAP standard no-charge license; see package notice and https://gsap.com/standard-license |
| howler | 2.2.4 | MIT |

| | | |
|---|---|---|
| msgpack-lite | 0.1.26 | MIT |
| normalize-wheel | 1.0.1 | BSD-3-Clause |
| pdf-parse | 2.4.5 | Apache-2.0 |
| pg | 8.23.1 | MIT |
| seedrandom | 3.0.5 | MIT |
| sharp | 0.34.5 | Apache-2.0 |
| stats-gl | 3.6.0 | MIT |

| Package | Locked version | License recorded in package/lock |
|---|---|---|
| stylus | 0.64.0 | MIT |
| three | 0.183.2 | MIT |
| tweakpane | 4.0.5 | MIT |
| uuid | 11.1.0 | MIT |
| vite | 7.2.4 | MIT |
| vite-plugin-node-polyfills | 0.24.0 | MIT |
| vite-plugin-restart | 2.0.0 | MIT |
| vite-plugin-top-level-await | 1.6.0 | MIT |
| vite-plugin-wasm | 3.5.0 | MIT |